Mining Silent Data Defects in LLM-Synthesized Agent Environments

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1. Problem and Motivation

Agent World Model (AWM) [1] automatically generates executable, database-backed environments for training and evaluating tool-using AI agents. Because these worlds are synthesized by large language models, their initial database states may contain data-quality defects. Some defects cause explicit SQL execution errors, but others execute successfully while creating implausible or inconsistent world states, such as unrealistic numerical values or inconsistent temporal relationships. These silent defects matter because agent tasks, tool interactions, and verification all depend on the generated database state.

This project asks:

Can data-mining methods detect SQL-valid but semantically corrupted records in LLM-synthesized agent environments, including defects that ordinary database execution does not detect?

The project first reproduces AWM's database initialization behavior, then extends it with a controlled benchmark of silent semantic corruptions.

2. Dataset

The project uses the publicly released Snowflake AgentWorldModel-1K dataset [2] and official AWM implementation [3], containing 1,000 environments with database schemas, sample data, tasks, tools, and environment code. Reproducing the official database-reset process yields 129,328 SQL `INSERT` statements, so *one instance = one original synthesized INSERT statement*, giving 129,328 instances.

For the extension, a selected subset of valid statements will be corrupted *in place*, keeping the total instance count at 129,328 rather than duplicating clean/corrupted copies. Since naturally occurring semantic errors lack ground-truth labels, controlled SQL-valid corruptions provide explicit labels: subtle numeric anomalies, temporally inconsistent values, contextually implausible values, and references to valid but semantically incorrect records.

3. Methods and Baselines

The reproduction stage characterizes AWM initialization failures using the official execution pipeline. The main extension compares three approaches for detecting silent semantic corruption:

- a simple deterministic rule-based baseline;
- Isolation Forest as an unsupervised anomaly detector trained on unmodified records; and
- Random Forest as a supervised classifier using the controlled corruption labels.

Features are derived only from information available before validation: SQL structure, column/value characteristics, schema metadata, numeric distributions, temporal information, and relational context.

Corruptions use multiple severity levels. Beyond obvious anomalies, the study includes plausible-looking errors, such as large but positive price shifts or small temporal reversals, so the task cannot be solved solely by trivial rules such as testing whether a value is negative.

4. Evaluation and Leakage Prevention

Statements from the same environment share schemas, generation context, and related data, so environments rather than individual statements are assigned to training and test partitions, preventing statements from the same generated world from appearing in both sets.

The primary metric is **PR-AUC**, with **Precision@K** and **Recall@K** measuring whether defects concentrate near the top of the anomaly ranking. Experiments will also compare performance across corruption types and severity, and a generalization experiment will hold out selected corruption types from training to test detection of unseen corruption types.

5. Feasibility

The AWM repository and dataset are publicly accessible and have already been downloaded and executed locally; the official initialization procedure has been reproduced across all 1,000 environments. A pilot on 22 environments confirmed that many deliberately corrupted values remain fully valid to SQLite while altering the semantic world state. The planned experiments use standard tabular and anomaly-detection algorithms and require no large-scale reinforcement learning, LLM training, or GPU compute, making the proposed reproduction, extension, and evaluation feasible within the semester.

References

[1] Z. Wang et al., "Agent World Model: Infinity Synthetic Environments for Agentic Reinforcement Learning," in *Proc. International Conference on Machine Learning (ICML)*, 2026.

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[5] L. Breiman, “Random Forests,” *Machine Learning*, vol. 45, pp. 5–32, 2001.